
ENGINEERING TRIPOS PART IIA

ELECTRICAL AND INFORMATION ENGINEERING TEACHING LABORATORY

MODULE EXPERIMENT 3F1

FLIGHT CONTROL

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Objectives:

- Simulation of various aircraft models on the computer.
- Study real-time (manual) control and the limitations imposed by time delays.
- Design of a simple autopilot.
- Illustrate frequency response concepts in analogue and digital control systems, conditions for oscillation in feedback systems and stability.
- Gain familiarity with Python and Jupyter Notebooks.

3F1 Lab Worksheet

2.1 Simplified Aircraft Model

$$\text{Transfer function} = \frac{10}{s^2 + 10s}$$
$$\text{num} = [10] \quad \text{den} = [1, 10, 0]$$

2.2 Modelling Manual Control

$$\text{Controller transfer function} = K(s) = ke^{-sD}$$

$$k = 0.8 \quad D = 1 \text{ s}$$

$$\text{Phase margin} = 45 \text{ degrees}$$

$$\text{Amount of extra time delay which can be tolerated} = \frac{45 \times \pi/180}{0.8} \approx 0.98 \text{ s (approx 1.0 s)}$$

2.3 Pilot Induced Oscillation

$$\text{Period of oscillation (observed)} = 5 \text{ s}$$

$$\text{Period of oscillation (theoretical)} = \frac{2\pi}{\omega_{pc}} = \frac{2\pi}{0.8} \approx 7.85 \text{ s}$$

2.4 Sinusoidal disturbances

$$\text{Maximum stabilising gain} = 4.5 \text{ (Gain Margin)}$$

$$\text{Gain at 0.66 Hz} = 5.5 \text{ dB}$$

$$\text{Phase at 0.66 Hz} = -90 \text{ degrees}$$

2.5 Unstable Aircraft

$$\text{Fastest pole at } T = 0.6$$

3.2 Autopilot with Proportional Control

$$\text{Proportional gain } K_c = 12$$

$$\text{Period of oscillation } T_c = 2 \text{ s}$$

3.3 Autopilot with PID Control

$$\text{Transfer function of PID controller} = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

$$\text{PID constants: } K_p = 7.2 \quad T_i = 1.0 \quad T_d = 0.25$$

$$\text{Adjusted value of } T_d = 0.35$$

3.4 Integrator Wind-up

$$\text{Integrator bound } Q = \frac{d \cdot T_i}{K_p} = \frac{2 \times 1.0}{7.2} \approx 0.278$$

3F1 Flight Control Lab Report

Question 1. (*§2.2 Modelling Manual Control*) Nyquist diagram (from Bode diagram) for controller in series with plant.

System Model: The simplified aircraft dynamics satisfy

$$\ddot{y}(t) + M\dot{y}(t) = Nx(t)$$

Taking the Laplace transform (with zero initial conditions), the plant transfer function is:

$$G(s) = \frac{N}{s^2 + Ms} = \frac{N}{s(s + M)}$$

This represents an integrator ($1/s$) in series with a first-order lag ($1/(s + M)$). The human controller is modelled as a proportional gain k with pure time delay D :

$$K(s) = ke^{-sD}$$

Open-Loop Transfer Function:

$$L(s) = K(s)G(s) = \frac{kN \cdot e^{-sD}}{s(s + M)}$$

Frequency Response Analysis: Substituting $s = j\omega$:

$$L(j\omega) = \frac{kN \cdot e^{-j\omega D}}{j\omega(j\omega + M)}$$

The denominator can be written as:

$$j\omega(j\omega + M) = j\omega \cdot |j\omega + M| \cdot e^{j \arctan(\omega/M)} = \omega \sqrt{\omega^2 + M^2} \cdot e^{j(\pi/2 + \arctan(\omega/M))}$$

Therefore, the magnitude and phase are:

$$|L(j\omega)| = \frac{kN}{\omega \sqrt{\omega^2 + M^2}}, \quad \angle L(j\omega) = -\frac{\pi}{2} - \arctan\left(\frac{\omega}{M}\right) - \omega D$$

Asymptotic Behaviour:

- **Low frequency** ($\omega \rightarrow 0$): $|L| \rightarrow \infty$, $\angle L \rightarrow -\pi/2$
- **High frequency** ($\omega \rightarrow \infty$): $|L| \rightarrow 0$, $\angle L \rightarrow -\infty$

Substituting experimental values ($M = N = 10$, $k = 0.8$, $D = 1$ s):

$$L(j\omega) = \frac{8e^{-j\omega}}{j\omega(j\omega + 10)}$$

From the Bode diagram (Figure 2), the gain crossover frequency $\omega_{gc} \approx 0.8$ rad/s. At this frequency:

$$\angle L(j \cdot 0.8) \approx -140^\circ$$

The phase margin is $\text{PM} = 180^\circ - 140^\circ = 40^\circ$ (approximately 45° from graph).

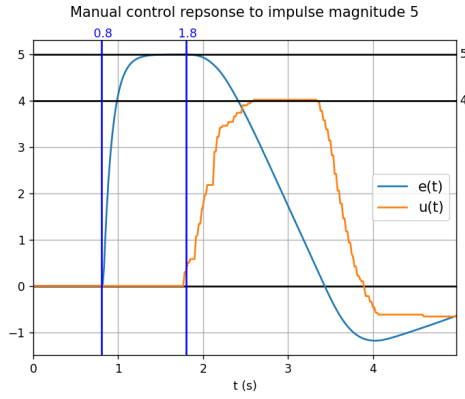


Figure 1: Manual control response to impulse disturbance. From the plot: $k \approx u_{max}/e_{max} = 4/5 = 0.8$, $D \approx 1$ s (delay before response).

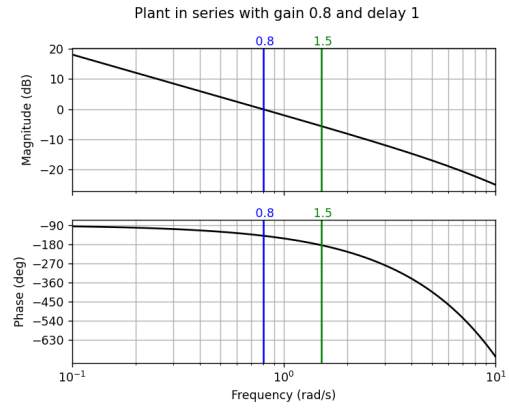
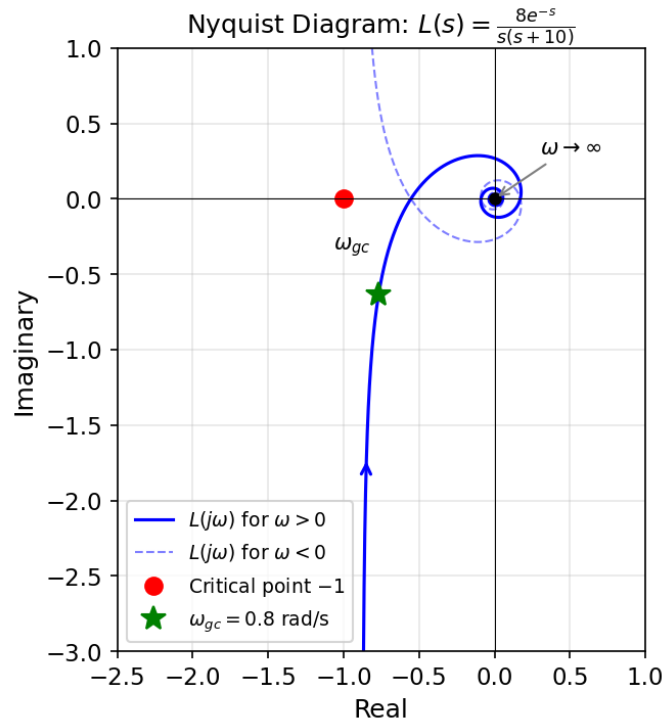


Figure 2: Bode diagram: open-loop $L(s)$ with $k = 0.8$, $D = 1$ s. Phase margin $\approx 45^\circ$ at $\omega_{gc} = 0.8$ rad/s.

Constructing Nyquist from Bode: The Nyquist plot traces $L(j\omega)$ as ω varies from 0 to ∞ . Starting from the negative imaginary axis (phase -90° , large magnitude), the locus spirals clockwise towards the origin. The critical point -1 is not encircled, confirming closed-loop stability.



Nyquist diagram of $L(s) = K(s)G(s)$. The locus spirals clockwise from the negative imaginary axis toward the origin. The critical point -1 (red) is not encircled.

Question 2. (§2.2 Modelling Manual Control) Are you using any integral action? Give a brief explanation. What does this imply about the accuracy of the model of the human controller?

Yes. In Figure 3, after the step disturbance, the error signal $e(t) \rightarrow 0$ while the control input $u(t)$ settles to a non-zero constant. For a purely proportional controller $u = ke$, this would be impossible—zero error implies zero output. The sustained $u \neq 0$ when $e = 0$ indicates integral action:

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau$$

The integral term accumulates past error, allowing $u_{ss} \neq 0$ even when $e_{ss} = 0$.

Model accuracy: The simple model $K(s) = ke^{-sD}$ is purely proportional and neglects this low-frequency integral behaviour. It captures fast reactions but not the pilot's adaptive error accumulation.

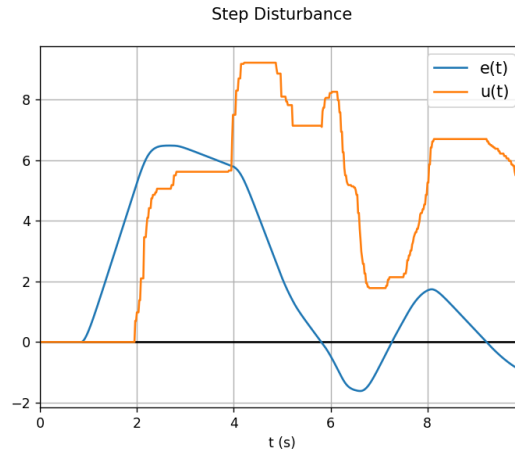


Figure 3: Step disturbance response showing integral action

Question 3. (§2.3 Pilot Induced Oscillation) Explain the oscillation of the feedback loop. How does your observed period of oscillation compare to the theoretical prediction?

PIO occurs when the pilot controller's time delay introduces sufficient phase lag that the total loop phase reaches -180° at a frequency where the loop gain exceeds unity. The PIO aircraft has $G_1(s) = c/(Ts + 1)^3$ which contributes up to -270° phase lag, plus the pilot delay e^{-sD} adds $-\omega D$ radians. At the phase crossover frequency ω_{pc} where $\angle L(j\omega_{pc}) = -180^\circ$, if $|L(j\omega_{pc})| > 1$, the system is unstable and oscillates at frequency ω_{pc} . From Bode (Figure 6), $\omega_{pc} \approx 0.8$ rad/s, giving theoretical period $T = 2\pi/\omega_{pc} \approx 7.85$ s. The observed period (≈ 5 s) is shorter, suggesting the pilot controller adapts their gain/delay under oscillatory conditions.

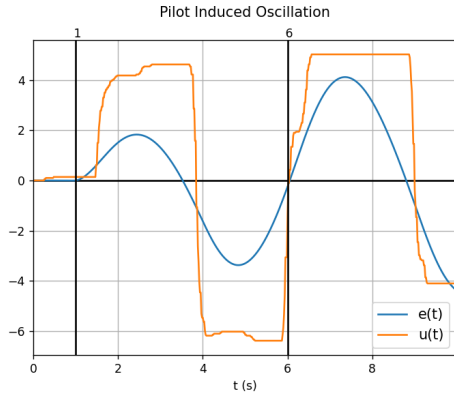


Figure 4: PIO with manual control. Period ≈ 5 s.

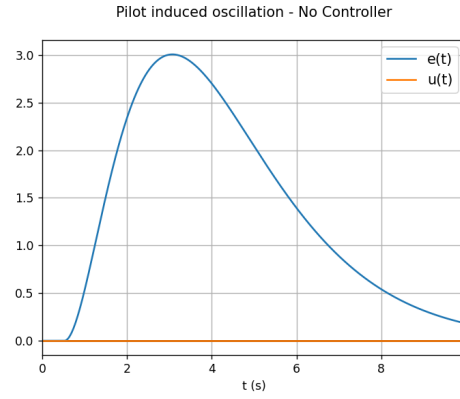


Figure 5: No control: aircraft is naturally stable.

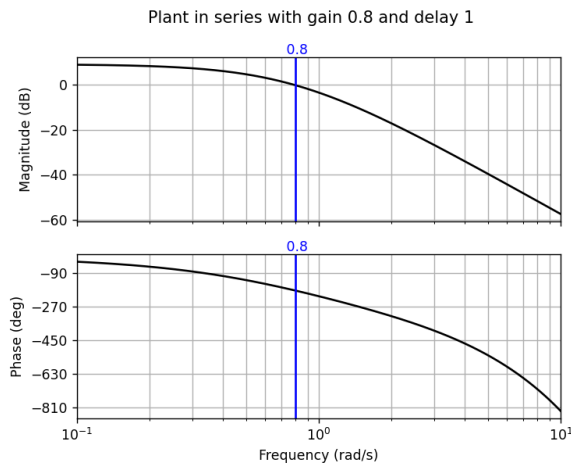


Figure 6: PIO Bode diagram. Phase crossover at $\omega_{pc} \approx 0.8$ rad/s.

Question 4. (§2.3 Pilot Induced Oscillation) Can you give a rough guideline to the control designer to make PIO less likely?

To prevent PIO, ensure adequate phase margin at frequencies where the pilot is active (< 1 Hz):

- Avoid plants with high phase lag at low frequencies
- Design for phase margin $> 45^\circ$ at the expected pilot crossover frequency
- Consider rate limiters to prevent high-frequency pilot inputs

Question 5. (§2.4 Sinusoidal Disturbances) Was your manual input able to reduce the error (as compared to providing no input)?

No. The closed-loop response to disturbance d is $y = \frac{G}{1 + KG}d$. For disturbance rejection, we need $|1 + KG| > 1$. At 0.66 Hz ($\omega = 4.15$ rad/s), pilot delay adds phase $-\omega D = -4.15 \times 1 \approx -238^\circ$. Combined with plant phase ($\approx -90^\circ$ from Bode), total phase approaches -330° , meaning KG points nearly opposite to -1 . This makes $|1 + KG| < 1$, so the disturbance is actually *amplified* rather than attenuated. Manual control worsens the oscillation.

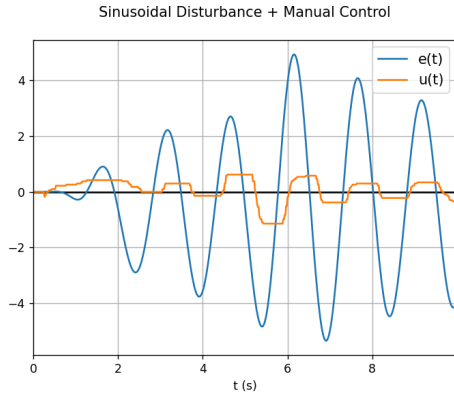


Figure 7: Sinusoidal + manual control.

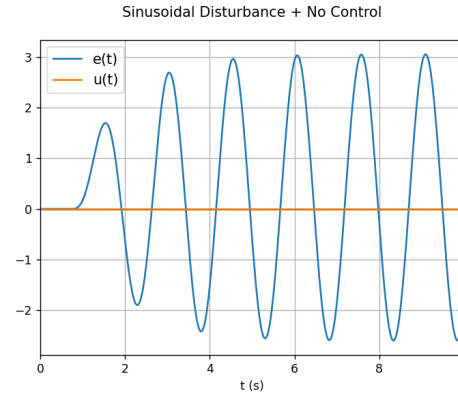


Figure 8: Sinusoidal + no control.

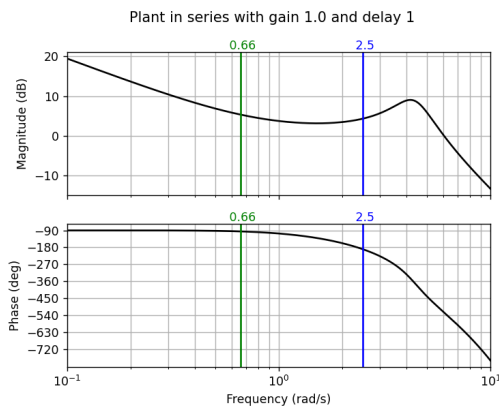


Figure 9: F4E Bode: gain margin 4.5; at 0.66 Hz: 5.5 dB, -90° .

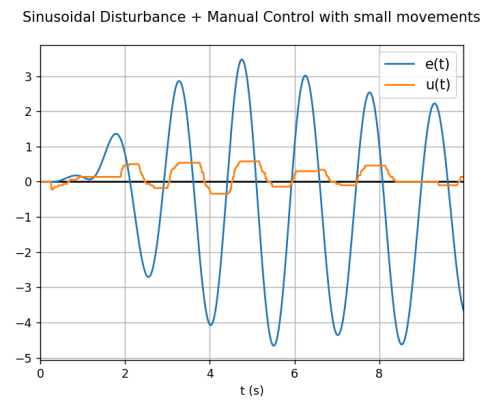


Figure 10: Small movements: still ineffective.

Question 6. (§2.5 *An Unstable Aircraft*) Nyquist diagram for $G_2(s)$.

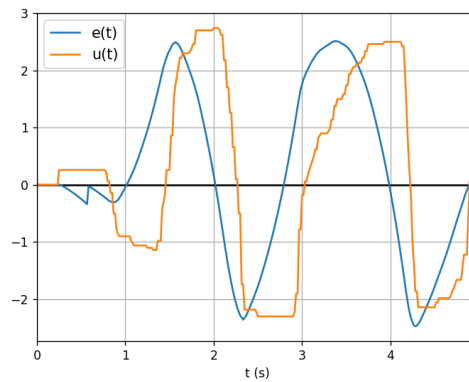


Figure 11: Unstable aircraft ($T = 0.6$) manually stabilized. Oscillatory near stability limit.

The unstable aircraft has transfer function:

$$G_2(s) = \frac{2}{sT - 1}$$

with one unstable (RHP) pole at $s = 1/T > 0$.

Frequency Response Derivation: Substituting $s = j\omega$:

$$G_2(j\omega) = \frac{2}{j\omega T - 1} = \frac{2}{-(1 - j\omega T)} = \frac{-2(1 + j\omega T)}{(1 - j\omega T)(1 + j\omega T)} = \frac{-2(1 + j\omega T)}{1 + \omega^2 T^2}$$

Therefore: $\text{Re}[G_2(j\omega)] = \frac{-2}{1 + \omega^2 T^2}$, $\text{Im}[G_2(j\omega)] = \frac{-2\omega T}{1 + \omega^2 T^2}$

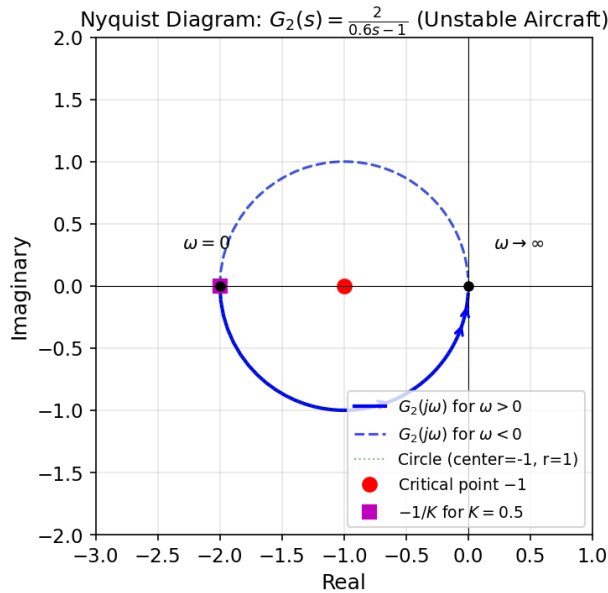
Geometric Interpretation: Letting $x = \text{Re}$ and $y = \text{Im}$:

$$(x + 1)^2 + y^2 = \left(\frac{-2}{1 + \omega^2 T^2} + 1 \right)^2 + \left(\frac{-2\omega T}{1 + \omega^2 T^2} \right)^2 = 1$$

This is a circle centered at $(-1, 0)$ with radius 1.

- At $\omega = 0$: $G_2(0) = -2$ (leftmost point on real axis)
- As $\omega \rightarrow \infty$: $G_2 \rightarrow 0$ (approaches origin)

The locus is traced clockwise as ω increases from 0 to ∞ .



Nyquist diagram of $G_2(s) = 2/(0.6s - 1)$. Circle centered at $(-1, 0)$ with radius 1.

Question 7. (§2.5 An Unstable Aircraft) Explain, using the Nyquist criterion, why the feedback system is stable with a proportional gain greater than 0.5.

Nyquist Criterion: For a system with P open-loop poles in the RHP, closed-loop stability requires $N = -P$ encirclements of $-1/K$, where $N > 0$ is counter-clockwise. Here $P = 1$ (one unstable pole), so we need $N = -1$: one *clockwise* encirclement of $-1/K$. The Nyquist locus is a circle from -2 to 0 on the real axis, traversed clockwise. For $-1/K$ to be encircled clockwise:

$$-2 < -\frac{1}{K} < 0 \implies 0 < \frac{1}{K} < 2 \implies K > 0.5$$

For $K > 0.5$, the point $-1/K$ lies inside the circle, giving the required clockwise encirclement for stability.

Question 8. (§2.5 An Unstable Aircraft) Sketch of a Nyquist diagram for $G_2(s)$ with a small time delay D .

With delay, the transfer function becomes $G_2(s)e^{-sD}$. The frequency response is:

$$G_2(j\omega)e^{-j\omega D} = \frac{2}{j\omega T - 1} \cdot e^{-j\omega D}$$

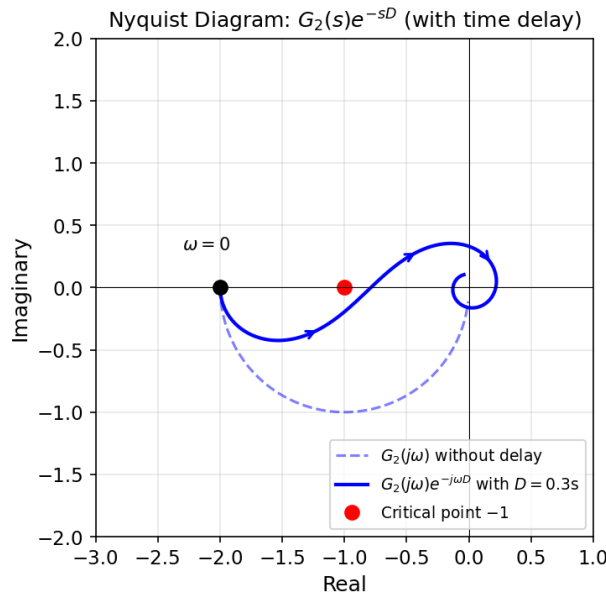
The delay contributes phase $-\omega D$ without changing magnitude:

$$|G_2(j\omega)e^{-j\omega D}| = |G_2(j\omega)|, \quad \angle(G_2e^{-j\omega D}) = \angle G_2 - \omega D$$

The effects on Nyquist locus:

- At $\omega = 0$: No change; still starts at $G_2(0) = -2$.
- As ω increases: Additional clockwise rotation by ωD radians.
- The circular locus becomes a spiral curving inward toward the origin.

For large D , the spiral may cross $-1/K$ multiple times, potentially destabilizing the system. This explains why manual stabilization becomes impossible when pilot delay D exceeds the system time constant T .



Nyquist diagram with time delay. Dashed: original $G_2(j\omega)$. Solid: with delay $e^{-j\omega D}$. The spiral deviates from the circle due to phase lag.

Question 9. (§3.4 Integrator Wind-up) Explain how you calculated the bound on Q .

PID Controller: $u(t) = K_p \left(e + \frac{1}{T_i} \int_0^t e \, d\tau + T_d \frac{de}{dt} \right)$

Steady-state analysis: After settling, $e_{ss} = 0$ and $\dot{e} = 0$. To cancel a step disturbance d , we need $u_{ss} = d$. Only the integral term contributes:

$$u_{ss} = \frac{K_p}{T_i} I_{ss} = d \implies I_{ss} = \frac{d \cdot T_i}{K_p}$$

Minimum bound: For $d = 2$, $T_i = 1.0$, $K_p = 7.2$:

$$Q = \frac{2 \times 1.0}{7.2} = \frac{5}{18} \approx 0.278$$

Setting Q smaller would prevent full disturbance rejection (steady-state error).

Setting Q larger allows more wind-up overshoot.

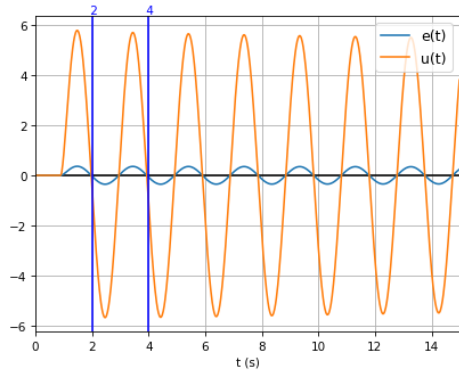


Figure 12: Proportional control: $K_c = 12$, $T_c = 2$ s.

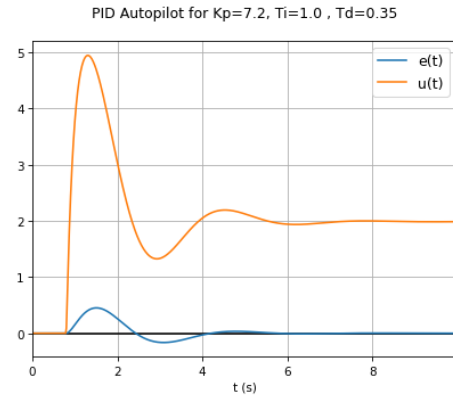


Figure 13: PID with $T_d = 0.35$: reduced oscillation.

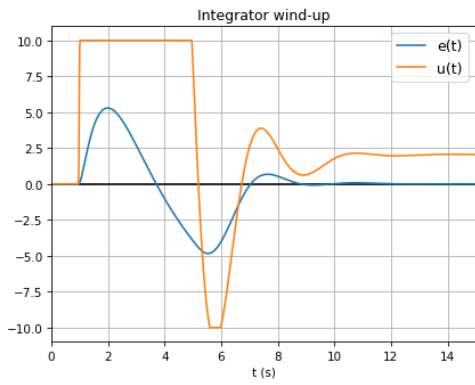


Figure 14: Integrator wind-up: large overshoot.

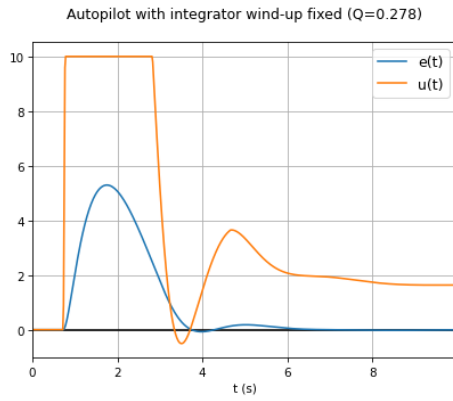


Figure 15: Wind-up fixed with $Q = 0.278$.